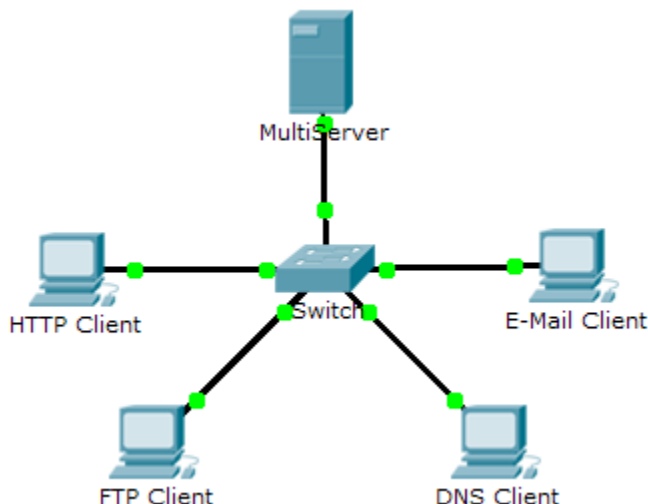


# Packet Tracer Simulation - TCP and UDP Communications

## Topology



## Objectives

**Part 1: Generate Network Traffic in Simulation Mode**

**Part 2: Examine the Functionality of the TCP and UDP Protocols**

## Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

## Part 1: Generate Network Traffic in Simulation Mode

### Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.
- Close the **MultiServer** window.

### Step 2: Generate web (HTTP) traffic.

- a. Switch to Simulation mode.
- b. Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.
- c. In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- d. Minimize, but do not close, the **HTTP Client** configuration window.

### Step 3: Generate FTP traffic.

- a. Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.
- c. Minimize, but do not close, the **FTP Client** configuration window.

### Step 4: Generate DNS traffic.

- a. Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.
- c. Minimize, but do not close, the **DNS Client** configuration window.

### Step 5: Generate Email traffic.

- a. Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.
- b. Click **Compose** and enter the following information:
  - 1) **To:** user@multiserver.pt.ptu
  - 2) **Subject:** Personalize the subject line
  - 3) **E-Mail Body:** Personalize the Email
- c. Click **Send**.
- d. Minimize, but do not close, the **E-Mail Client** configuration window.

### Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

## Part 2: Examine Functionality of the TCP and UDP Protocols

### Step 1: Examine multiplexing as all of the traffic crosses the network.

You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

- a. Click **Capture/Forward** once. All of the PDUs are transferred to the switch.
- b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?  
The PDUs are stored in the switch.
- c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?  
Multiplexing

- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?

Different protocols have different colours

- e. Click **Back** eight times. This should reset the simulation.

**Note:** Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

### Step 2: Examine HTTP traffic as the clients communicate with the server.

- a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:
  - 1) Click **Edit Filters** and toggle the **Show All/None** check box.
  - 2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.
- b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?  
TCP  
  
Are these communications considered to be reliable?  
Yes
- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?  
1025, 80, 0, 0. Flags: SYN
- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?  
80, 1025, 0, 1. Flags: ACK + SYN.  
The source port and the destination ports are reversed. The acknowledgment number is set to 1.
- g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.
- h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?  
1025, 80, 1, 1. Flags: PSH + ACK. The source port and the destination ports are reversed. The sequence number and the acknowledgment number are set to 1.
- i. Click **Back** until the simulation is reset.

### Step 3: Examine FTP traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.
- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?  
TCP

Are these communications considered to be reliable?

Yes

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?  
1025, 21, 0, 0. Flags: SYN
- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?  
21, 1025, 0, 1. Flags: ACK + SYN.  
The source port and the destination ports are reversed. The acknowledgment number is set to 1.
- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?  
1025, 21, 1, 1. Flags: ACK. The source port and the destination ports are reversed. The sequence number and the acknowledgment number are set to 1.
- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.
- i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?  
Welcome to PT Ftp server
- j. Click **Back** until the simulation is reset.

### Step 4: Examine DNS traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.
- b. Click the PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?  
UDP, Not reliable  
  
Are these communications considered to be reliable?
- d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?  
1025, 53. UDP does not need to establish a reliable connection.
- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?  
53, 1025. The source port and the destination ports are reversed.
- g. What is the last section of the **PDU** called?  
DNS Answer
- h. Click **Back** until the simulation is reset.

### Step 5: Examine email traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **POP3**, **SMTP** and **TCP**.

- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?  
TCP  
  
Are these communications considered to be reliable?  
Yes
- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?  
1025, 25, 0, 0. Flag: SYN
- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?  
25, 1025, 0, 1. Flags: ACK + SYN.  
The source port and the destination ports are reversed. The acknowledgment number is set to 1.
- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?  
1025, 21, 1, 1. Flags: ACK. The source port and the destination ports are reversed. The sequence number and the acknowledgment number are set to 1.
- h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.
- i. How are the port and sequence numbers different from the previous two **PDU**s?  
1025, 80, 1, 1. Flags: PSH + ACK. The source port and the destination ports are reversed. The sequence number and the acknowledgment number are set to 1.
- j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?  
Port 25: SMTP    Port 110: POP3
- k. Click **Back** until the simulation is reset.

### Step 6: Examine the use of port numbers from the server.

- a. To see TCP active sessions, perform the following steps in quick succession:
  - 1) Switch back to **Realtime** mode.
  - 2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **netstat** command. What protocols are listed in the left column? TCP  
What port numbers are being used by the server?  
21
- c. What states are the sessions in?  
ESTABLISHED
- d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open? FTP

Why doesn't this session close like the other three? (Hint: Check the minimized clients)  
The server is waiting for a username

### Suggested Scoring Rubric

| Activity Section                                           | Question Location | Possible Points | Earned Points |
|------------------------------------------------------------|-------------------|-----------------|---------------|
| Part 2: Examine Functionality of the TCP and UDP Protocols | Step 1            | 15              |               |
|                                                            | Step 2            | 15              |               |
|                                                            | Step 3            | 15              |               |
|                                                            | Step 4            | 15              |               |
|                                                            | Step 5            | 15              |               |
|                                                            | Step 6            | 25              |               |
| Total Score                                                |                   | 100             |               |